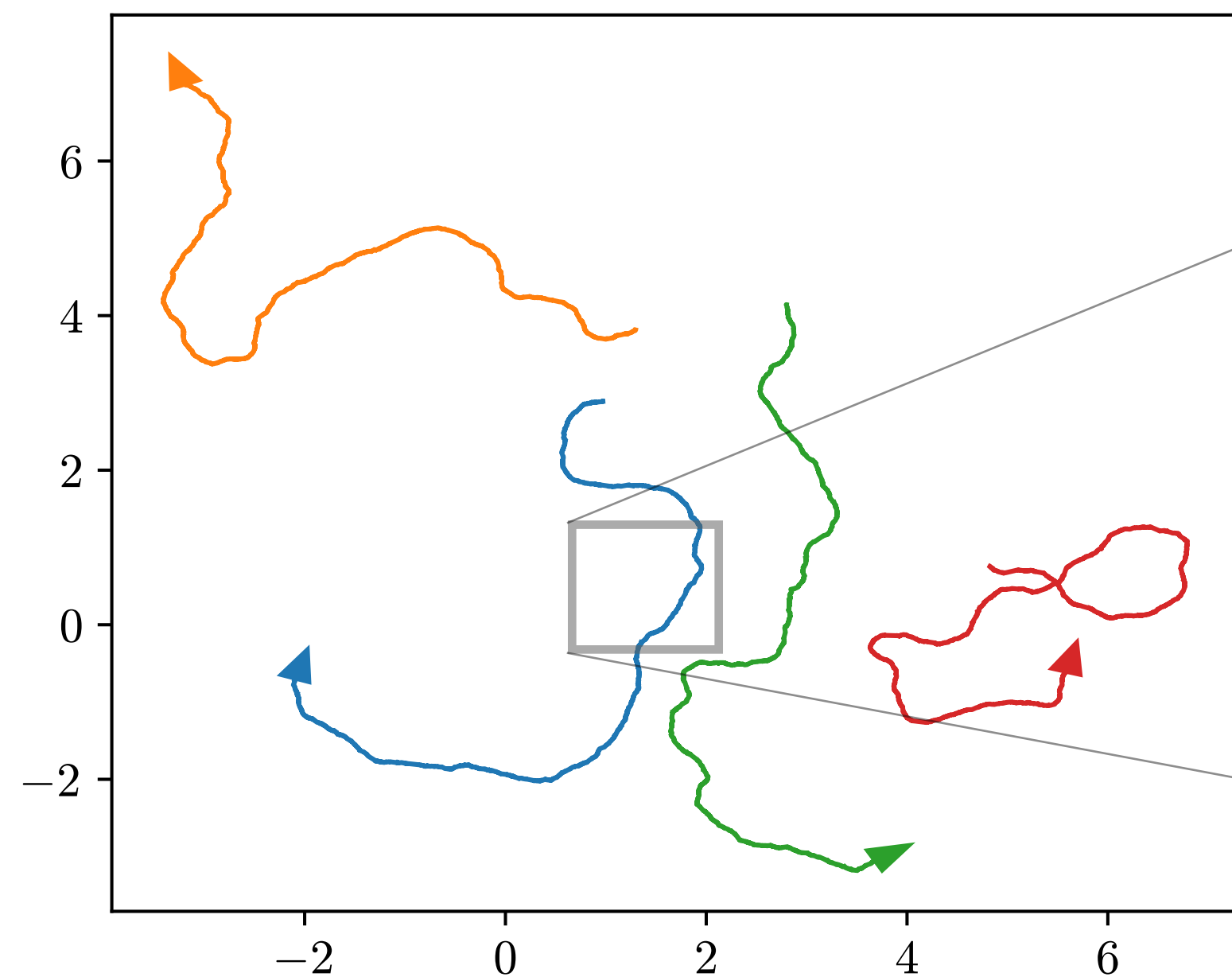


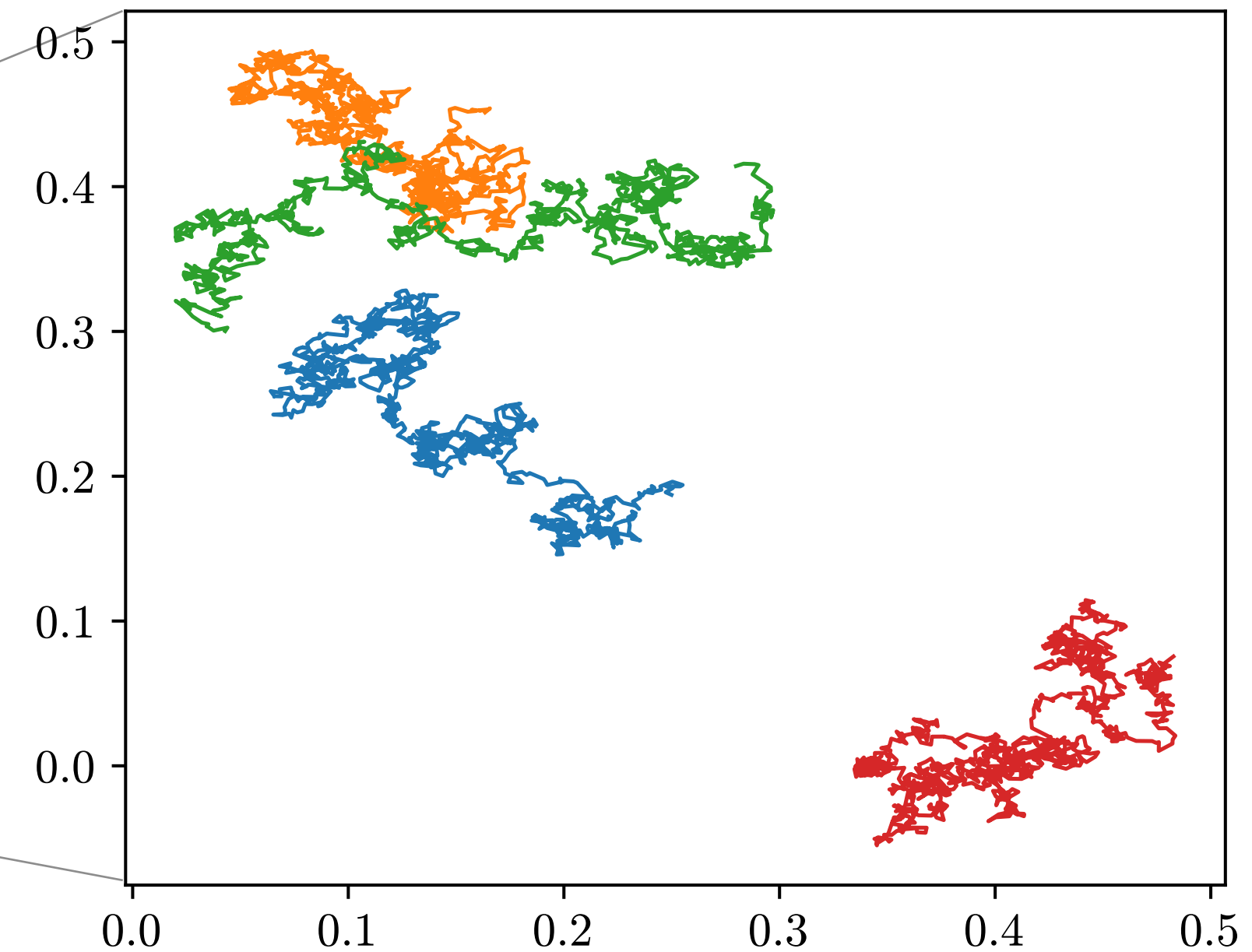
Active Brownian particles

- 2D Active Brownian particles

$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \sqrt{2} dB_t^i \end{cases}$$



$\text{Pe} = 2, \sigma_x = 10^{-3}$



$\text{Pe} = 0, \sigma_x = 10^{-3}$

Active Vlasov interaction

$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \chi F\left((X_t^i, \theta_t^i), \mu_t^N\right)dt + \sqrt{2}dB_t^i \end{cases}$$

- Interaction strength $\chi > 0$
- Empirical measure: $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{(X_t^i, \theta_t^i)}$, McKean-Vlasov